Factor analysis

The orthogonal factor model

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Factor analysis (FA)

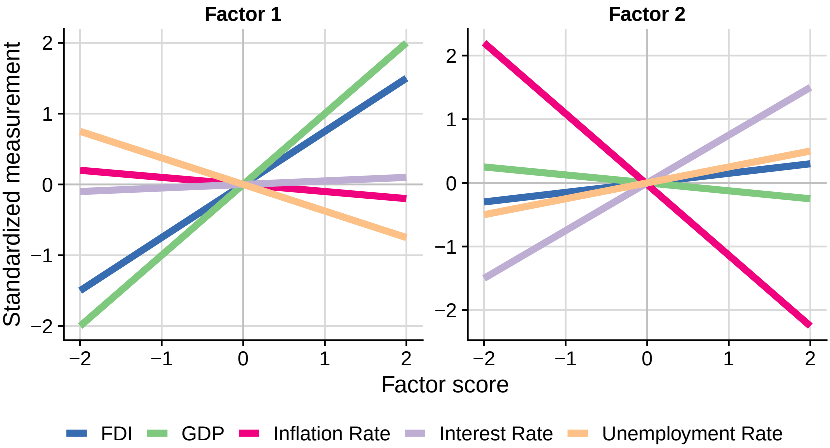
- Context: $\mathbf{X} \sim\ ?$
  - Goal: Explain covariance structure of $\mathbf{X}$ in terms of a smaller number of unobserved variables

Economic example of factor analysis

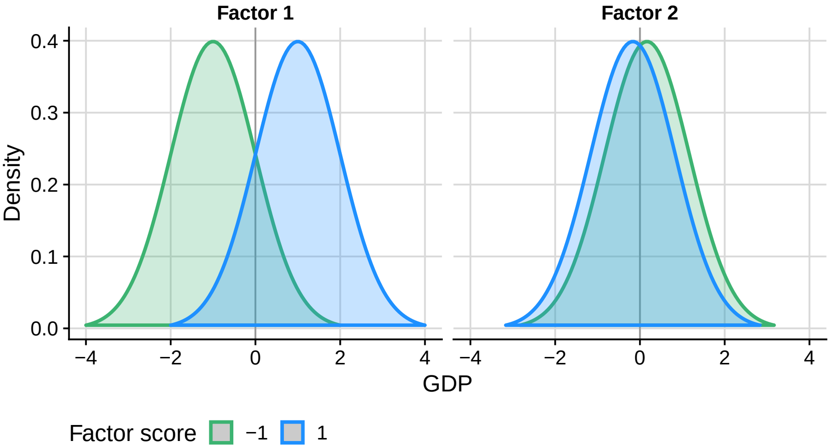
Suppose an economist wants to understand the factors that influence a country's economic growth. They might collect data on various macroeconomic indicators such as:

- **Gross Domestic Product (GDP)**: A measure of the economic output of a country.
- **Inflation Rate**: The rate at which the general level of prices for goods and services is rising.
- **Unemployment Rate**: The percentage of the labor force that is jobless and actively looking for employment.
- **Interest Rates**: The amount charged by lenders to borrowers for the use of assets, expressed as a percentage of the principal.
- **Foreign Direct Investment (FDI)**: Investment made by a firm or individual in one country into business interests located in another country.

Average levels of economic variables versus factors (cont.)



Distribution of GDP given factors



- $\mathrm{E}[X_{\mathrm{GDP}}] = f_1 - \frac{1}{6} f_2$

Sources of randomness

Error terms

- Given the factor scores for a given observation, the observed variables are not determined - there is some variation around that value.
- For example, in the previous slide GDP had a distribution given the factors.
- So, two countries with the same levels for factors 1 and 2 might have different GDPs.

Factors

- Factors are random variables (albeit unobserved), with a new draw for each observation.
- For example, one country will randomly have factor scores of 1 and -1.5, whereas another would have factor scores of -0.5 and -1.2.

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- Both error terms and factors typically have a mean of zero, but different approaches to factor analysis will place different assumptions regarding their variances and, especially, their covariances and distributional shape.

Other examples

- **Psychological Assessments**
  - *Observed*: Responses to psychological questionnaires.
  - *Hidden*: Personality traits (e.g., extraversion, neuroticism).
- **Consumer Behavior**
  - *Observed*: Shopping patterns, demographic data.
  - *Hidden*: Purchase drivers (e.g., parental status, age, income).
- **Medical Diagnosis**
  - *Observed*: Transcriptomic profiles.
  - *Hidden*: Disease states (e.g., cured, infected, advanced).

The orthogonal factor model (OFM)

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- We'll discuss the following:
  - Assumptions
  - Model specification
  - Methods for estimation
    - Principal component method
    - Maximum likelihood
  - Factor rotation
  - Factor scores

Assumptions

Linearity

- The relationship between the observed variables and the factors is linear.
- For example, the relationship between GDP and the factors is linear (e.g., $\mathrm{E}[X_{\mathrm{GDP}}] = f_1 - \frac{1}{6} f_2$).

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Independence

- The factors are uncorrelated with each other.
  - For example, the first factor being being high does not imply the second factor is (more likely to be) high.
- The factors are uncorrelated with the error terms.
  - For example, the first factor being high does not imply the error (change from the mean) for GDP is high.

The orthogonal factor model

Let

- $\mathbf{X} : p \times 1$ be the random vector of observed variables,
- $\mathbf{F} : m \times 1$ be the random vector of unobserved factors,
- $\boldsymbol{\epsilon} : p \times 1$ be the random vector of unobserved error terms, and
- $\mathbf{L} : p \times m$ be a matrix of constants,

satisfying

$$\mathbf{X} - \boldsymbol{\mu} = \mathbf{L}\mathbf{F} + \boldsymbol{\epsilon},$$

where

- $\boldsymbol{\mu}$ is the mean of $\mathbf{X}$,
- $\mathrm{E}[\mathbf{F}] = \mathbf{0}$,
- $\mathrm{Var}[\mathbf{F}] = \mathbf{I}$,
- $\mathrm{E}[\boldsymbol{\epsilon}] = \mathbf{0}$,
- $\mathrm{Var}[\boldsymbol{\epsilon}] = \boldsymbol{\Psi}$, a diagonal matrix, and
- $\mathrm{Cov}[\mathbf{F}, \boldsymbol{\epsilon}] = \mathbf{0}$.

Example

- Let $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$, $\mathbf{F} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$, and $\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix}$.
- Then, the model can be written as

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix}.$$

- For X_1, this implies that

$$\begin{aligned} X_1 - \mu_1 &= l_{11}F_1 + l_{12}F_2 + \epsilon_1 \\ X_1 &= \mu_1 + l_{11}F_1 + l_{12}F_2 + \epsilon_1. \end{aligned}$$

- Regarding the covariances, for example we have that

$$\mathrm{Cov}[F_1, F_2] = 0,$$

- and that

$$\mathrm{Cov}[F_1, \epsilon_1] = 0.$$

Example (cont.)

- We have that

$$\mathrm{Var}[F_1] = \mathrm{Var}[F_2] = 1,$$

- and that

$$\mathrm{Var}[\epsilon_1] = \psi_1, \quad \mathrm{Var}[\epsilon_2] = \psi_2.$$

- As such, the factors have unit variance, whereas the errors have variable-dependent variance.

- The factor scores are therefore unit-free, indicating the direction and magnitude in a standardised manner.
  - For example, in the economic example a large positive second first factor level indicates above-average economic activity.
- The errors therefore help provide the correct "scaling".
  - For example, interest rates and GDP will have different variabilities (without standardisation), which will (partially) be accounted for by differing error term variances.

Terminal terminology

- F_j: j-th common factor
  - "Common" in the sense of contributing to all the observed variables
- ϵ_i: i-th specific factor
  - "Specific" in the sense of contributing to only one observed variable
- l_{ij}: loading of i-th variable on j-th common factor
  - The extent to which the j-th factor influences the i-th observed variable

Source of correlation between observed variables

- In the OFM, the error terms are uncorrelated, the factor terms are uncorrelated and the factors are uncorrelated with the error terms.
- How do we then get correlation between the observed variables?

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- In the OFM, the error terms are uncorrelated, the factor terms are uncorrelated and the factors are uncorrelated with the error terms.
- How do we then get correlation between the observed variables?
- The same factor contributing to two observed variables will induce correlation between them.
- For example, if there is only one factor and so $\mathrm{E}[X_1] = l_{11}F_1$ and $\mathrm{E}[X_2] = l_{21}F_1$, then as long as l_{11} and l_{21} are both non-zero, then X_1 and X_2 will be correlated.
- This is the essence of factor analysis: to explain the correlation between observed variables in terms of a smaller number of unobserved factors.
- We'll now get a formula for the covariance between observed variables in terms of the factor loadings.

Covariance between observed variables

- The OFM model specification implies the following formula for the covariance between the observed variables (proofs at end of the slides):

$$\mathrm{Cov}[\mathbf{X}] = \mathbf{L}\mathbf{L}' + \mathbf{\Psi}$$

- For a given observed variable, we have that

$$\mathrm{Var}[\mathbf{X}_i] = \sum_{j=1}^{m} l_{ij}^2 + \psi_{ii}.$$

- For a pair of observed variables, we have that

$$\mathrm{Cov}[X_i, X_j] = \sum_{k=1}^{m} l_{ik} l_{jk}.$$

- The diagonal elements are the specific variances, and "inflate" the variances of the observed variables (over and above that implied by the loadings).
- As $\boldsymbol{\psi}$ is diagonal, the off-diagonal elements of the covariance matrix are determined by the factor loadings.

Accuracy of implied covariance matrix

- The covariance matrix between the observed variables has $p(p+1)/2$ unique elements, and is typically full rank.
- The factor model implies that the covariance matrix is equal to

$$\mathbf{L}\mathbf{L}' + \mathbf{\Psi}.$$

- Whilst Ψ is full rank, $\mathbf{L}\mathbf{L}'$ is rank $m \leq p$ and is fully responsible for the off-diagonal covariances.
- We can always perfectly reproduce the diagonal elements of the covariance matrix when $m = p$ (as many factors as observed variables) by setting

$$\mathbf{L} = \mathbf{V}\mathbf{D}/\sqrt{n-1},$$

- where $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}'$.
- However, we almost always want fewer factors than observed variables, and so we will not be able to perfectly reproduce the off-diagonal elements of the covariance matrix.
- In practice, no problem - an approximation is fine.

Covariance between observed and unobserved variables

- By the assumptions of the OFM, we have that

$$\mathrm{Cov}[\mathbf{X}, \mathbf{F}] = \mathbf{L}.$$

- For a specific observed variable and factor, we have that

$$\mathrm{Cov}[X_i, F_j] = l_{ij}.$$

- As such, the covariance between the observed variables and the factors is determined by the factor loadings.

Solutions are not unique

- The solution to an OFM model consists of the loadings matrix $\mathbf{L}$ and the error variances $\boldsymbol{\Psi}$.
- Two solutions are equivalent if a) meet the assumptions of the OFM and b) imply same covariance matrix and specific variances.
- In the OFM model, an orthonormal rotation of the loading matrix can yield the same covariance matrix and specific variances.

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- Two solutions are equivalent if a) meet the assumptions of the OFM and b) imply same covariance matrix and specific variances.
- In the OFM model, an orthonormal rotation of the loading matrix can yield the same covariance matrix and specific variances.
- Suppose that $\mathbf{T}$ is an $m \times m$ orthonormal matrix, i.e. $\mathbf{T}\mathbf{T}' = \mathbf{I}_m$.
- Then, if $\mathbf{L}$ is a solution to the OFM, so is $\mathbf{L}^* = \mathbf{L}\mathbf{T}$.
- In other words, the solutions $\mathbf{L}$

$$\mathbf{X} = \mathbf{L}\mathbf{F} + \boldsymbol{\epsilon}$$

- and $\mathbf{L}\mathbf{T} = \mathbf{L}^*$

$$\mathbf{X} = (\mathbf{L}\mathbf{T})(\mathbf{T}'\mathbf{F}) + \boldsymbol{\epsilon} = \mathbf{L}^*\mathbf{F}^* + \boldsymbol{\epsilon}$$

- are equivalent.

Demonstrating non-uniqueness

- Firstly, we show that the mean and variance assumptions are met for the common factors:

$$\mathrm{E}[\mathbf{F}^*] = \mathrm{E}[\mathbf{T}'\mathbf{F}] = \mathbf{T}'\mathrm{E}[\mathbf{F}] = 0, \text{ and}$$
$$\mathrm{Cov}[\mathbf{F}^*] = \mathrm{T}'\mathrm{Cov}[\mathbf{F}]\mathrm{T} = \mathrm{T}'\mathrm{T} = \mathrm{I}.$$

- Secondly, we show that the specific factors are unchanged:

$$\begin{aligned} \mathbf{X} - \boldsymbol{\mu} &= \mathbf{L}\mathbf{F} + \boldsymbol{\epsilon} \\ &= \mathbf{L}\mathbf{T}\mathbf{T}'\mathbf{F} + \boldsymbol{\epsilon} \text{ (as } \mathbf{T}\mathbf{T}' = \mathbf{I}\text{), and} \\ &= \mathbf{L}^*\mathbf{F}^* + \boldsymbol{\epsilon}, \end{aligned}$$

where $\mathbf{L}^* = \mathbf{L}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}'\mathbf{F}$.

- Thirdly, we show that the estimated covariance matrix also does not change:

$$\begin{aligned} \hat{\mathrm{Var}}[\mathbf{X}] &= \mathbf{L}\mathbf{L}' + \boldsymbol{\Psi}, \\ &= \mathbf{L}\mathbf{T}\mathbf{T}'\mathbf{L}' + \hat{\boldsymbol{\Psi}}, \\ &= \mathbf{L}^*\mathbf{L}^{*\prime} + \boldsymbol{\Psi}. \end{aligned}$$

Exploiting the lack of uniqueness

- Since the solution to the OFM is not unique, we can find initial values for $\hat{\mathbf{L}}$ and then rotate them to find a solution that is more interpretable.
- By "interpretable", we mean one where each factor contributes prominently to a minority of the variables, e.g. factor 1 contributes strongly only to variables 1 and 2, factor 2 contributes strongly only to variables 2 and 5, etc.

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- Since the solution to the OFM is not unique, we can find initial values for $\hat{\mathbf{L}}$ and then rotate them to find a solution that is more interpretable.
- By "interpretable", we mean one where each factor contributes prominently to a minority of the variables, e.g. factor 1 contributes strongly only to variables 1 and 2, factor 2 contributes strongly only to variables 2 and 5, etc.

- In the next couple of slides, we'll consider two approaches for fitting the factor analysis model to a given dataset, and then we'll consider rotating those initial solutions.

First method of estimation: Principal component method I

The spectral decomposition of the covariance matrix is

$$\begin{aligned}
\boldsymbol{\Sigma}_{p\times p} &= \lambda_1 \boldsymbol{e_1 e_1'} + \lambda_2 \boldsymbol{e_2 e_2'} + \cdots + \lambda_m \boldsymbol{e_m e_m'} \\
&= \left[\sqrt{\lambda_1} e_1 \quad \sqrt{\lambda_2} e_2 \quad \dots \quad \sqrt{\lambda_m} e_m\right] \begin{bmatrix} \sqrt{\lambda_1} e_1 \\ \sqrt{\lambda_2} e_2 \\ \vdots \\ \sqrt{\lambda_m} e_m \end{bmatrix} \\
&= \boldsymbol{L}_{p\times m} \boldsymbol{L}'_{p\times m}
\end{aligned}$$

where $m \le p$ refers to the number of factors.

Thus the matrix of estimated factor loadings $\{\tilde{l}_{ij}\}$ is given by the eigenvalue-eigenvector pairs of the sample covariance matrix, $\boldsymbol{S}$,

$$\tilde{\boldsymbol{L}} = \left[\sqrt{\hat{\lambda}_1 \hat{e}_1} \quad \sqrt{\hat{\lambda}_2 \hat{e}_2} \quad \dots \quad \sqrt{\hat{\lambda}_m \hat{e}_m}\right].$$

If we allow for specific factors, we can add $\boldsymbol{\Psi} = diag(\Psi_i)$ and estimate these by the diagonal elements of

$$\boldsymbol{S} - \tilde{\boldsymbol{L}}\tilde{\boldsymbol{L}}' = s_{ii} - \sum_{j=1}^{m} \tilde{l}_{ij}^2 = s_{ii} - \tilde{h}_i^2.$$

First method of estimation: Principal component method II

If we standardize the variables by subtracting the sample mean and dividing by the sample standard deviation, $z_{ji} = \frac{x_{ji} - \bar{x}_i}{\sqrt{s_{ii}}}$, we work with the sample correlation matrix, $\boldsymbol{R}$.

This is called the "Principal Component Solution" because the factor loadings are scaled coefficients of the first few sample principal components.

Further note:

- The estimated loadings for a given factor do not change as the number of factors is increased.

  So if $m = 1, \tilde{\boldsymbol{L}} = \left[\sqrt{\hat{\lambda}_1}\hat{e}_1\right]$ and if $m = 2, \tilde{\boldsymbol{L}} = \left[\sqrt{\hat{\lambda}_1}\hat{e}_1 \quad \sqrt{\hat{\lambda}_2}\hat{e}_2\right], etc.$

  Where it is exactly the same value for $\sqrt{\hat{\lambda}_1}\hat{e}_1$ in both solutions.

- It can be shown that the contribution to the total sample variance, $s_{11} + s_{22} + \cdots + s_{pp} = tr(\boldsymbol{S})$, from the first common factor is $\tilde{l}_{11}^2 + \tilde{l}_{21}^2 + \cdots + \tilde{l}_{p1}^2 = \left(\sqrt{\hat{\lambda}_1}\hat{\boldsymbol{e}}_1\right)'\left(\sqrt{\hat{\lambda}_1}\hat{\boldsymbol{e}}_1\right) = \hat{\lambda}_1$. Thus the proportion of total variance due to the j^{th} factor is

$$\frac{\hat{\lambda}_j}{s_{11}+s_{22}+\cdots+s_{pp}} \text{ for a factor analysis of } \boldsymbol{S}$$

and

$$\frac{\hat{\lambda}_j}{p} \text{ for a factor analysis of } \boldsymbol{R}.$$

For p standardized variables the sum of the variances = p.

Beyond that the explained variance does not increase

Thus we can take m to equal the number of eigenvalues of **R** greater than 1 or the number of positive eigenvalues of **S**.

Example 1: Fitting using PC method I

A consumer preference study where customers were asked to rate a new product in terms of
(i)Taste, (ii) Good buy for money, (iii) Flavor, (iv) Suitable for snack, (v) Provides lots of energy using a 7-point scale.

In R:

```
> #Read in correlation matrix

> corr<-matrix(c(1,0.02,0.96,0.42,0.01,0.02,1,0.13,0.71,
+                    0.85,0.96,0.13,1,0.50,0.11,
+                    0.42,0.71,0.50,1,0.79,0.01,0.85,0.11,0.79,1),5,5,byrow=T)

> corr
     [,1] [,2] [,3] [,4] [,5]
[1,] 1.00 0.02 0.96 0.42 0.01
[2,] 0.02 1.00 0.13 0.71 0.85
[3,] 0.96 0.13 1.00 0.50 0.11
[4,] 0.42 0.71 0.50 1.00 0.79
[5,] 0.01 0.85 0.11 0.79 1.00
```

We observe high correlations among some of the attributes, for example between [1]+[3] and [2]+[5]+[4]. This indicates the possibility of finding latent groups, and hence a successful factor analysis

Example 1: Fitting using PC method II

```
> #Perform eigenvalue decomposition of correlation matrix
> eigencorr<-eigen(corr)
> eigencorr$values
[1] 2.85309042 1.80633245 0.20449022 0.10240947 0.03367744
> eigencorr$vectors
          [,1]        [,2]        [,3]       [,4]         [,5]
[1,] 0.3314539 -0.60721643  0.09848524  0.1386643  0.701783012
[2,] 0.4601593  0.39003172  0.74256408 -0.2821170  0.071674637
[3,] 0.3820572 -0.55650828  0.16840896  0.1170037 -0.708716714
[4,] 0.5559769  0.07806457 -0.60158211 -0.5682357  0.001656352
[5,] 0.4725608  0.40418799 -0.22053713  0.7513990  0.009012569

> #Factor loadings are scaled eigenvectors
> l<-eigencorr$vectors[,1:2]%*%diag(sqrt(eigencorr$values[1:2]))
> l
          [,1]       [,2]
[1,] 0.5598618 -0.8160981
[2,] 0.7772594  0.5242021
[3,] 0.6453364 -0.7479464
[4,] 0.9391057  0.1049187
[5,] 0.7982069  0.5432281
> #Communalities=sum of squared loadings
> h<-diag(l%*%t(l))
> h
[1] 0.9794614 0.8789200 0.9758829 0.8929275 0.9322311
```

(2.85+1.81)/5=0.93. Thus first 2 common factors will account for 93% of total variance.

5=sum of variances for standardized variables

$\tilde{\boldsymbol{L}} = \left[\sqrt{\hat{\lambda}_1}\hat{e}_1 \quad \sqrt{\hat{\lambda}_2}\hat{e}_2 \quad \cdots \quad \sqrt{\hat{\lambda}_m}\hat{e}_m\right]$

Based on $m = 2$

Based on size and sign of loadings, 1st factor a “general factor” whilst 2nd factor groups [1]+[3] and contrasts to rest.
1st Factor could also be viewed as grouping [2]+[4]+[5]

(i)Taste,
(ii) Good buy for money,
(iii) Flavor,
(iv) Suitable for snack,
(v) Provides lots of energy

$\boldsymbol{LL}'=$
$h_i^2 = l_{i1}^2 + l_{i2}^2 + \cdots + l_{im}^2$

Example 1: Fitting using PC method III

```
> #Calculate specific variances
> psi<-1-h
```

$S = \tilde{L}\tilde{L}' = s_{ii} - \sum_{j=1}^{m} \tilde{l}_{ij}^2$

$= s_{ii} - \tilde{h}_i^2$

and for correlation matrix

$s_{ii} = 1$

```
> psi
[1] 0.02053865 0.12107998 0.02411712 0.10707250 0.06776888


> #Create diagonal matrix of specific variances
> psi_mat<-diag(psi)
> psi_mat
           [,1]    [,2]       [,3]      [,4]       [,5]
[1,] 0.02053865 0.00000 0.00000000 0.0000000 0.00000000
[2,] 0.00000000 0.12108 0.00000000 0.0000000 0.00000000
[3,] 0.00000000 0.00000 0.02411712 0.0000000 0.00000000
[4,] 0.00000000 0.00000 0.00000000 0.1070725 0.00000000
[5,] 0.00000000 0.00000 0.00000000 0.0000000 0.06776888
```

$LL' + \Psi$

```
> #Re-create correlation matrix
> reproduced_corr<-l%*%t(l)+psi_mat
> reproduced_corr
            [,1]        [,2]      [,3]      [,4]        [,5]
[1,] 1.000000000 0.007357539 0.9716968 0.4401455 0.003558179
[2,] 0.007357539 1.000000000 0.1095187 0.7849273 0.905175175
[3,] 0.971696840 0.109518709 1.0000000 0.5275656 0.108806485
[4,] 0.440145533 0.784927343 0.5275656 1.0000000 0.806595488
[5,] 0.003558179 0.905175175 0.1088065 0.8065955 1.000000000
```

Difference between reproduced correlation matrix and observed correlation matrix is an indication of how good an approximation was achieved by using just 2 factors.

```
> #as compared to observed correlation matrix:
> corr
     [,1] [,2] [,3] [,4] [,5]
[1,] 1.00 0.02 0.96 0.42 0.01
[2,] 0.02 1.00 0.13 0.71 0.85
[3,] 0.96 0.13 1.00 0.50 0.11
[4,] 0.42 0.71 0.50 1.00 0.79
[5,] 0.01 0.85 0.11 0.79 1.00
```

Example 2: Fitting using PC method I

Weekly rates of return for p=5 stocks, (i) JP Morgan, (ii) Citibank, (iii) Wells Fargo, (iv) Royal Dutch Shell, (v) ExxonMobil for n=103 successive weeks.

```
> #Read in correlation matrix
> corr<-matrix(c(1,0.632,0.511,0.115,0.155,0.632,1,0.574,
+                   0.322,0.213,0.511,0.574,1,
+                   0.183,0.146,0.115,0.322,0.183,1,0.683,0.155,0.213,0.146,
+                   0.683,1),5,5,byrow=T)
> corr
      [,1]  [,2]  [,3]  [,4]  [,5]
[1,] 1.000 0.632 0.511 0.115 0.155
[2,] 0.632 1.000 0.574 0.322 0.213
[3,] 0.511 0.574 1.000 0.183 0.146
[4,] 0.115 0.322 0.183 1.000 0.683
[5,] 0.155 0.213 0.146 0.683 1.000

> #Perform eigenvalue decomposition of correlation matrix
> eigencorr<-eigen(corr)
> eigencorr$values
[1] 2.4376148 1.4062612 0.5001857 0.4000328 0.2559055
> eigencorr$vectors
           [,1]       [,2]        [,3]       [,4]        [,5]
[1,] -0.4692143  0.3677864 -0.60456518  0.3618911  0.38484653
[2,] -0.5322254  0.2364426 -0.13554332 -0.6305079 -0.49489270
[3,] -0.4652234  0.3154110  0.77108235  0.2911524  0.06888643
[4,] -0.3873348 -0.5850179  0.09596914 -0.3800925  0.59501717
[5,] -0.3607117 -0.6058861 -0.11113451  0.4921936 -0.49818028
```

First 2 components account for (2.44+1.41)/5=0.77 (77%) of total variability

First component roughly a weighted average of all 5 stocks, thus a general market component. Second component seems to contrast JP Morgan+Citibank+Wells Fargo (banking stocks) to Shell+Exxon(oil stocks).

Example 2: Fitting using PC method II

```
> #Factor loadings are scaled eigenvectors
> #First using just one factor
> eigencorr$values[1]
[1] 2.437615
> l<-
eigencorr$vectors[,1:1]*(sqrt(eigencorr$values[1]))

> l
[1] -0.7325779 -0.8309562 -0.7263470 -0.6047405 -
0.5631743
```

First component of 2-dim solutions same as one-dim solution

```
> #Communalities=sum of squared loadings
> h<-diag(l%*%t(l))
> h
[1] 0.5366703 0.6904882 0.5275799 0.3657111 0.3171653
> #Calculate specific variances
> psi<-1-h
> psi
[1] 0.4633297 0.3095118 0.4724201 0.6342889 0.6828347
```

```
> #Using two factors

> eigencorr$values[1:2]
[1] 2.437615 1.406261
> scale2<-diag(sqrt(eigencorr$values[1:2]),2,2)
> scale2
         [,1]     [,2]
[1,] 1.561286 0.000000
[2,] 0.000000 1.185859
>
> l<-eigencorr$vectors[,1:2]%*%scale2
> l
           [,1]       [,2]
[1,] -0.7325779  0.4361428
[2,] -0.8309562  0.2803875
[3,] -0.7263470  0.3740330
[4,] -0.6047405 -0.6937486
[5,] -0.5631743 -0.7184954
>
> #Communalities=sum of squared loadings
> h<-diag(l%*%t(l))
> h
[1] 0.7268909 0.7691054 0.6674806 0.8469982 0.8334010
> #Calculate specific variances
> psi<-1-h
> psi
[1] 0.2731091 0.2308946 0.3325194 0.1530018 0.1665990
```

Larger unique variances for banking stocks than for oil stocks. Oil stock more strongly correlated and hence easier to explain by common factor

Second method of estimation: Maximum likelihood

If the common factors $\boldsymbol{F}$, and the specific factors, $\boldsymbol{\epsilon}$, are normally distributed, then the maximum likelihood estimates of the factor loadings and the specific variances may be obtained.

If the $\boldsymbol{F}_j$ and $\boldsymbol{\epsilon}_j$ are jointly normally distributed, the observations

$$\boldsymbol{X}_j - \boldsymbol{\mu} = \boldsymbol{L}\boldsymbol{F}_j + \boldsymbol{\epsilon}_j \sim \text{Normal}.$$

And the likelihood, $L(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ depends on L and Ψ through $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}' + \boldsymbol{\Psi}$. This is not well-defined because of multiplicity of choices for L and hence we impose a uniqueness condition, $\boldsymbol{L}'\boldsymbol{\Psi}^{-1}\boldsymbol{L} = \boldsymbol{\Delta}$, a diagonal matrix.
The mle's of $\hat{\boldsymbol{L}}$ and $\hat{\boldsymbol{\Psi}}$ are obtained by numerical maximization.

Everything else remains as for the principal component estimates. Hence
Given mle estimates, $\hat{\boldsymbol{L}}$ and $\hat{\boldsymbol{\Psi}}$,

$$\hat{\boldsymbol{\Sigma}} = \hat{\boldsymbol{L}}\hat{\boldsymbol{L}}' + \hat{\boldsymbol{\Psi}},$$
$$\hat{h}_i^2 = \hat{l}_{i1}^2 + \hat{l}_{i2}^2 + \cdots + \hat{l}_{im}^2.$$

The residual covariance matrix is $\boldsymbol{\Sigma} - \hat{\boldsymbol{L}}\hat{\boldsymbol{L}}' - \hat{\boldsymbol{\Psi}}$.

Proportion of total sample variance due to j^{th} factor

$$= \frac{\hat{l}_{1j}^2 + \hat{l}_{2j}^2 + \cdots + \hat{l}_{pj}^2}{s_{11} + s_{22} + \cdots + s_{pp}}.$$

Example 3: Fitting using maximum likelihood I

```
> FA<-factanal(covmat=corr,correlation,factors=2,
+ rotation="none")
> FA

Call:
factanal(x = correlation, factors = 2, covmat =
corr, rotation = "none")

Uniquenesses:
[1] 0.416 0.276 0.541 0.005 0.530

Loadings:
     Factor1 Factor2
[1,]  0.121   0.755
[2,]  0.328   0.785
[3,]  0.188   0.650
[4,]  0.997
[5,]  0.685

               Factor1 Factor2
SS loadings      1.622   1.609
Proportion Var   0.324   0.322
Cumulative Var   0.324   0.646

The degrees of freedom for the model is 1 and the
fit was 0.0199
```

The factanal function in R uses mle estimation

Default output omits printing of very small loadings

```
> print(FA,cutoff=0)

Call:
factanal(x = correlation, factors = 2,
covmat = corr, rotation = "none")

Uniquenesses:
[1] 0.416 0.276 0.541 0.005 0.530

Loadings:
     Factor1 Factor2
[1,]  0.121   0.755
[2,]  0.328   0.785
[3,]  0.188   0.650
[4,]  0.997  -0.007
[5,]  0.685   0.027

               Factor1 Factor2
SS loadings      1.622   1.609
Proportion Var   0.324   0.322
Cumulative Var   0.324   0.646

The degrees of freedom for the model is 1
and the fit was 0.0199
```

To print all loadings set cut-off to 0.

Revisiting rotation

```
> FA1<-
factanal(covmat=corr,factors=2,rotation="none")
> FA1

Call:
factanal(factors = 2, covmat = corr, rotation =
"none")

Uniquenesses:
[1] 0.510 0.594 0.644 0.377 0.431 0.628

Loadings:
     Factor1 Factor2
[1,]  0.553   0.429
[2,]  0.568   0.288
[3,]  0.392   0.450
[4,]  0.740  -0.273
[5,]  0.724  -0.211
[6,]  0.595  -0.132

               Factor1 Factor2
SS loadings      2.209   0.606
Proportion Var   0.368   0.101
Cumulative Var   0.368   0.469

The degrees of freedom for the model is 4 and the fit
was 0.0109
```

Factor 1 does not distinguish between different subjects, while factor 2 seems to contrast 1+2=3 to 4+5+6, Humanities versus Mathematics.

The varimax rotation is an orthogonal rotation aimed at maximizing the variances of the squares of the factor loadings,

$$V = \frac{1}{p}\sum_{j=1}^{m}\left[\sum_{i=1}^{p} \hat{l}_{ij}^{*4} - (\sum_{i=1}^{p} \hat{l}_{ij}^{*2})^2/p\right].$$

```
> varimax(loadings(FA1))
$loadings

Loadings:
     Factor1 Factor2
[1,] 0.235   0.659
[2,] 0.323   0.549
[3,]         0.590
[4,] 0.771   0.170
[5,] 0.724   0.213
[6,] 0.572   0.210

               Factor1 Factor2
SS loadings      1.612   1.203
Proportion Var   0.269   0.201
Cumulative Var   0.269   0.469

$rotmat
           [,1]      [,2]
[1,]  0.8420397 0.5394156
[2,] -0.5394156 0.8420397
```

Factor 1 = 4+5=6, i.e., Mathematics and Factor 2=1+2=3, i.e., Humanities

This is rotation matrix T, L*=LT. For example 0.553*0.842+ 0.429*-0.539 =0.235

Factor scores

The estimated values of the common factors are called "factor scores", i.e., for the j^{th}case,

$\hat{\boldsymbol{f}}_j$ =estimated values attained by $\boldsymbol{F}_j$.

These can be estimated using a method of weighted least squares:

From the factor model, $\boldsymbol{X} - \boldsymbol{\mu} = \boldsymbol{LF} + \boldsymbol{\epsilon}$, we regard the specific factors,ϵ_i, as error terms that do not need to have equal variances and hence we use the reciprocal of their variances,$1/\Psi_i$, as weights in a weighted least squares minimization of

$$\sum_{i=1}^{p} \frac{e_i^2}{\Psi_i} = \boldsymbol{\epsilon}'\boldsymbol{\Psi}^{-1}\boldsymbol{\epsilon} = (\boldsymbol{x} - \boldsymbol{u} - \boldsymbol{Lf})'\Psi^{-1}(\boldsymbol{x} - \boldsymbol{u} - \boldsymbol{Lf}),$$

So in this equation, f are the variables to be estimated and all else are taken as known data values

where we treat the estimates loadings $(\boldsymbol{L})$, the specific variances $(\boldsymbol{\Psi})$ and the sample means (μ) as if they were true (as opposed to estimated) values.

This leads to $\hat{\boldsymbol{f}} = (\boldsymbol{L}'\boldsymbol{\Psi}^{-1}\boldsymbol{L})^{-1}\boldsymbol{L}'\boldsymbol{\Psi}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}) = (\hat{\boldsymbol{L}}'\hat{\boldsymbol{\Psi}}^{-1}\hat{\boldsymbol{L}})^{-1}\hat{\boldsymbol{L}}'\hat{\boldsymbol{\Psi}}^{-1}(\boldsymbol{x} - \overline{\boldsymbol{x}})$.

If $\hat{L}$ and $\hat{\Psi}$ are estimated using mle, they have to satisfy the uniqueness condition that $\hat{\boldsymbol{L}}'\hat{\boldsymbol{\Psi}}^{-1}\hat{\boldsymbol{L}} = \hat{\boldsymbol{\Delta}}$, **a** diagonal matrix.

If the correlation matrix is factored, we work with standardized variables, $\boldsymbol{Z} = \boldsymbol{V}^{-1}(\boldsymbol{X} - \boldsymbol{\mu})$.

Example 4: Factor scores

Example: Back to Factor Analysis of Stock Price Data

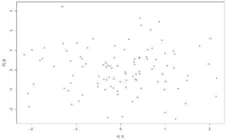
Need to read in actual data since we need X to calculate scores

```
> StockPriceDAta=read.table(file.choose())

> dimnames(StockPriceDAta) <- list (1:nrow(StockPriceDAta),
+c("JPMorgan","Citibank","WellsFargo","Shell","Exxon"))
> head(StockPriceDAta)
    JPMorgan   Citibank WellsFargo      Shell      Exxon
1  0.0130338 -0.0078431 -0.0031889 -0.0447693  0.0052151
2  0.0084862  0.0166886 -0.0062100  0.0119560  0.0134890
3 -0.0179153 -0.0086393  0.0100360  0.0000000 -0.0061428
4  0.0215589 -0.0034858  0.0174353 -0.0285917 -0.0069534
5  0.0108225  0.0037167 -0.0101345  0.0291900  0.0409751
6  0.0101713 -0.0121978 -0.0083768  0.0137083  0.0029895
> FA<-factanal(x=StockPriceDAta,correlation,factors=2,
+             rotation="none",scores="Bartlett")
> FA
Call:
factanal(x = StockPriceDAta, factors = 2, data =
correlation,     scores = "Bartlett", rotation = "none")
Uniquenesses:
  JPMorgan   Citibank WellsFargo      Shell      Exxon
     0.417      0.275      0.542      0.005      0.530
Loadings:
           Factor1 Factor2
JPMorgan    0.121   0.754
Citibank    0.328   0.786
WellsFargo  0.188   0.650
Shell       0.997
Exxon       0.685
```

```
> FA$scores
       Factor1     Factor2
1   -1.81015485  0.47157713
2    0.29926209  0.41007876
3   -0.15535353 -0.46550636
4   -1.20954479  0.95126882
5    0.94005033 -0.21296820
6    0.35335600 -0.51696625
7    0.99041738  0.84578680
8   0.08906855  1.03310622
…
97    1.50896270 -0.29191830
98    0.18010412  1.56056119
99   -0.22426808 -0.85235425
100 -0.19809233  0.08757334
101 -0.74631464  1.12899771
102 -0.24666460  0.20246081
103 -0.34665021 -1.01009961
> F<-matrix(FA$scores,103,2)
➤ plot(F[,1],F[,2])
```

So 103 sets of values for 2 new latent score variables, as opposed to 5 original stick returns. First latent factor represent oil returns and second latent factor represents banking returns. We can plot these latent variables (see below) or regress them as functions of other variables, anything you would normally do to variables.



Two factors are independent of one another hence the random scatter.

Take-aways I

- Conceptual understanding of factor analysis' relationship to PCA and multivariate regression
- Assumed mean and covariance structure of an orthogonal factor model
  - Implied form for covariance matrix
  - Communality and specific factors
  - Correlation between observed and unobserved variables
- Two methods of fitting the OFM to a given dataset
  - Principal component method (first principles)
  - Maximum likelihood (`factanal` package)

Take-aways II

- Factor rotations
  - Theory behind why they're permitted
  - Able to justify why they may be useful
- Interpretation of results of factor analysis
- Calculation of factor scores
  - From first principles

Resources

- Source code is on GitHub.
  - Suppressed code (e.g. to create figures) is available there.

References

Johnson, Richard A., and Dean W. Wichern. 2007. *Applied Multivariate Statistical Analysis*. 6th ed. Upper Saddle River, NJ: Pearson Education Inc.